RESEARCH IN SCIENCE & TECHNOLOGICAL EDUCATION

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NEW ARTICLE

Enhancing project-based chemistry learning in rural schools through virtual and augmented reality: evidence from Kazakhstan

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WHAT DID THE RESEARCHERS INVESTIGATE?

- Inadequate access to laboratory resources in remote secondary schools hinders efficient chemistry learning.
- The study investigates the effectiveness of integrating virtual and augmented reality (VR/AR) into project-based chemistry learning.
- In a quasi-experiment, 32 upper-secondary students engaged in VR/AR-supported environmental chemistry projects, while 32 received conventional instruction.

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WHAT DID THEY FIND?

- The experimental group showed substantially greater achievement gains than the control group (mean gains of 19 versus 3).
- Experimental students also showed increases in motivation and perceived understanding, two measures that were strongly positively related.
- The authors suggest VR/AR-supported projects may be a promising approach for chemistry education in resource-constrained rural contexts.

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